Gondwana Research 28 (2015) 391–406

Contents lists available at ScienceDirect

Gondwana Research

journal homepage: www.elsevier.com/locate/gr

$^{40}Ar/^{39}Ar$ geochronology and the paleoposition of Christmas Island (Australia), Northeast Indian Ocean

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ARTICLE INFO

Article history:
Received 14 October 2013
Received in revised form 3 April 2014
Accepted 3 April 2014
Available online 27 April 2014

Handling Editor: J.G. Meert

Keywords:
Paleomagnetism
Seamounts
Seismic tomography
Plate reconstruction
Indian Ocean
Capricorn plate

ABSTRACT

The Christmas Island Seamount Province is an extensive zone of volcanism in the Northeast Indian Ocean, consisting of numerous submerged seamounts and flat-topped guyots. Within this region lies two subaerial island groups, Christmas Island, and the Cocos Keeling archipelago. Christmas Island has experienced multiple episodes of volcanism that are exposed sporadically along its coastline. Here, we dated these volcanics using $^{40}Ar/^{39}Ar$ geochronology and analysed them for paleomagnetism. The oldest exposed volcanism occurred in the Eocene between 43 and 37 Ma. This is followed by a time gap of ~33 million years, before the eruption of a younger episode of Pliocene age (4.32 ± 0.17 Ma). It has, however, been suggested by previous workers that there is a much older Late Cretaceous event beneath the limestone which is unexposed. In addition, this study conducted the first paleomagnetic analysis of samples from Christmas Island to determine its paleoposition and the paleomagnetic polarity of the sampled sites. Two normal and two reversal magnetic events have been recorded, that agree with the geomagnetic reversal timescales. Late Eocene (38–39 Ma) palaeomagnetic data suggest a palaeolatitude of $-43.5^{\circ}{}^{+9.0^{\circ}}_{-11.2^{\circ}}$, which is further south than palaeolatitudes (around 30° S) derived from existing plate reconstruction models for the Australian plate. However, the Late Eocene palaeomagnetic data are limited (only two sites) and secular variation may not have been averaged out. During the Pliocene (ca. 4 Ma) we estimate a palaeolatitude of approximately 13° S. The presence of the Late Eocene ages at Christmas Island correlates well with the cessation of spreading of the Wharton Ridge (~43 Ma), the initiation of spreading along the South East Indian Ridge, and the transit of Christmas Island over a broad low velocity zone in the upper mantle. This suggests that changes in stress regimes following the tectonic reorganisation of the region (prior to~43 Ma) may have allowed deeper-origin mantle melts to rise. Similarly, changes in the plate's stress regime at the flexural bulge of the Sunda–Java subduction zone may be implicated in renewed melting at ~4 Ma, suggesting that tectonic stresses have exerted a first-order effect on the timing and emplacement of volcanism at Christmas Island.

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1. Introduction

1.1. Christmas Island Seamount Province (CHRISP)

The Northeast Indian Ocean is an understudied area, but the occurrence of large intraplate and subduction-zone earthquakes in the last ten years within the intraplate and subduction region of the Indo-Australian plate and a recent cruise has led to a re-emergence of geological interest (Abercrombie et al., 2003; DeMets and Royer, 2003; Hoernle et al., 2011; Hall, 2012; Yue et al., 2012). The Northeast Indian Ocean region is dominated by numerous submerged seamounts and two ocean islands whose origin has not received much attention until recently (Hoernle et al., 2011). The distribution of volcanism does not follow a linear hotspot model (Morgan, 1971). The region is dominated by numerous north–south trending fracture zones, but again, these do not seem to be exerting a fundamental control on the distribution of seamount volcanism, which predominantly trends east–west.

The Northeast Indian Ocean is divided into various physiographic divisions as shown in Fig. 1 comprising; The North West Shelf, Exmouth Plateau, Wombat Plateau, Argo Abyssal Plain, Cuvier Abyssal Plain, Gascoyne Abyssal Plain, and Platypus Spur. The North West Shelf is a continental margin extending from northern Australia to the Exmouth Plateau off the west coast that was formed due to multiple episodes of rifting and removal of continental fragments following the disintegration of Gondwana in the Late Cretaceous (Stagg et al., 1999; Heine and Müller, 2005). The Argo Abyssal Plain is located west of the North West Shelf in the easternmost corner of the Indian Ocean. It is

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http://dx.doi.org/10.1016/j.gr.2014.04.004
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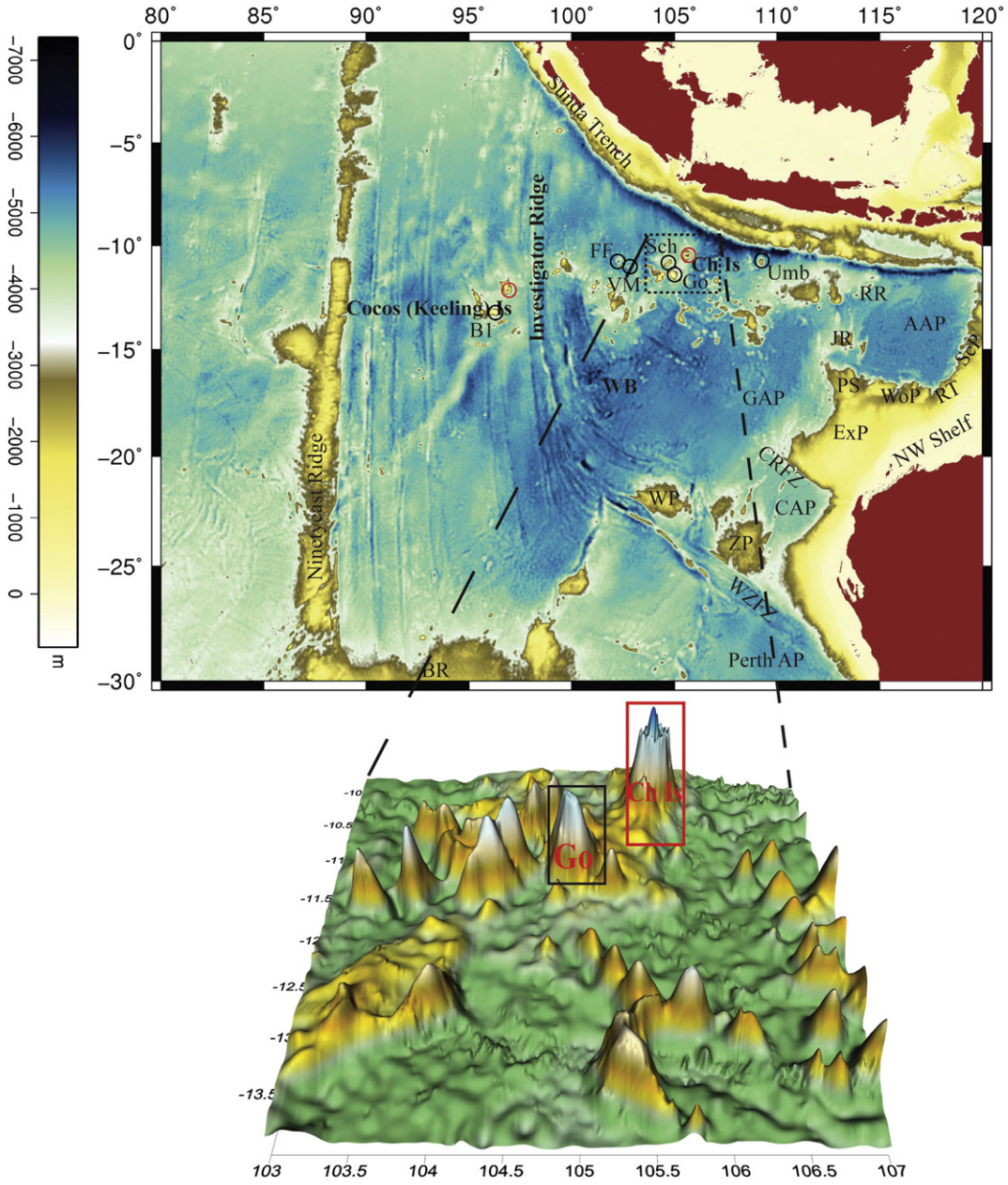


Fig. 1. Regional bathymetry map of Northeast Indian Ocean showing various tectonic features and the two islands (red circle) within the region. AAP, Argo Abyssal Plain; GAP, Gascoyne Abyssal Plain; CAP, Cuvier Abyssal Plain; Perth AP, Perth Abyssal Plain; RR, Roo Rise; JR, Joey Rise; ExP, Exmouth Plateau ScP, Scott Plateau; WoP, Wombat Plateau; PS, Platypus Spur; RT, Rowley Terrace; WP, Wallaby Plateau; ZP, Zenith Plateau; CRFZ, Cape Range Fracture Zone, WZFZ, Wallaby–Zenith Fracture Zone; WB, Wharton Basin; BR, Broken Ridge; Ch Is, Christmas Island and Go, Golden Bo'Sunbird (85 Ma) seamount are represented in red and black rectangle. Other seamounts are FF, Flying Fish (70 Ma); VM, Vening Meinesz (72 Ma); Sch, Shcherbakov (~81 Ma); and Umb, Umbgrove (no age available).

surrounded by the Java Trench in the north, and the submerged continental crust of the Scott Plateau in the northeast, the Rowley Terrace in the east and the Wombat Plateaus in the south. The Exmouth Plateau lies further south of Wombat Plateau. The western edges of the Argo Abyssal Plain are separated by the Joey and Roo Rises north of Platypus Spur, and by the Gascoyne Abyssal Plain in the west. The Gascoyne Abyssal Plain is bounded by the Exmouth Plateau in the west, Cuvier Abyssal Plain in the south and the Java Trench in the north, with Argo Abyssal Plain in the east and Wharton Basin and Christmas Islands in the west. Investigator Ridge, a roughly 1800 km long ridge trending north south along 98° E longitude is a distinctive feature of the Indo-Australian plate. It can be seen prominently in bathymetric maps as a continuous linear ridge that extends from 7°–18° S, beyond which it becomes patchy and broken before subducting underneath Sumatra as evident from Fig. 1.

The most important characteristic features of the Northeast Indian Ocean are the numerous submerged seamounts, volcanic guyots and two sub-aerial islands. The Christmas Island Seamount Province (CHRISP) lies within the Northeast Indian Ocean, south of the Java Trench and between the Ninetyeast Ridge and the North West Shelf off the west coast of Australia (Fig. 1). These volcanic edifices rise up to 3000 m below sea-level in a region where the abyssal plain depth ranges between 5500 and 6000 m. Most of these volcanic features occur within the latitudes 10° S and 15° S as a wide zone of seamounts rather than a linear volcanic chain. The two sub aerially exposed island groups are the Cocos (Keeling) Archipelago, and Christmas Island. Cocos

(Keeling) Islands are a coral atoll archipelago consisting of 27 islands. There are no volcanic exposures on the islands (Woodroffe et al., 1991; Woodroffe and Falkland, 1997). Hoernle et al. (2011) dredged volcanic rocks at depths of 2400–3100 m that yielded Late Palaeocene ages of 55.6 ± 0.2 Ma and 56.0 ± 0.2 Ma.

Christmas Island, on the other hand, is the sole island to have recorded sub-aerial intraplate volcanism, in the form of basaltic rocks that are exposed in some places on the island. Little published literature exists concerning Christmas Island's geology. Prominent amongst these are the works done by Andrews (1900) and Trueman (1965) on the geology of the island. Barrie (1967) discussed the phosphates of the island with minor volcanic examination based essentially on visual and field observations. Earlier works on Christmas Island by Rivereau (1965), Polak (1976), Pettifer and Polak (1979), and Barrie (1967) were primarily geomorphological and geophysical surveys. Barrie (1967), by visual examination of volcanic rock and its breakdown into soil, and faunal assemblages within interbedded limestones suggested, that the main phase of volcanism ceased in the Early Tertiary and the island became volcanically extinct in the Pliocene. The presence of these volcanic outcrops, therefore, makes this island significant in constraining the tectonic and volcanic evolution of the region.

There is a report of paleomagnetic analysis carried out at Christmas Island by the Bureau of Mineral Resources. This preliminary study was based on one sample per site to understand rock properties and mentioned some locations for detailed paleomagnetic investigations (Polak, 1976). We conduct a detailed high resolution paleomagnetic analyses using larger sample size to determine the paleomagnetic record of Christmas Island, the only volcanically exposed representative of submarine volcanism in the CHRISP region.

Hoernle et al. (2011) has put forward a mechanism describing the origin of the CHRIS Province. They proposed shallow recycling of continental lithosphere as a mechanism for volcanic generation of the CHRISP. They argue that decompressional melting of volatile rich Archean SCLM and MORB source mantle leads to the formation of CHRISP near the MOR during the dispersal of Gondwana. In addition, Hoernle et al. (2011) suggested that the volcanic province was formed near the mid-oceanic ridge. They carried out $^{40}Ar/^{39}Ar$ age dating on dredged samples within the intraplate region, and found that they get younger towards the west. Ages between 136.2 ± 0.17 Ma in the Argo Abyssal Plain, and 47.0 ± 0.2 Ma in the east, south of Cocos (Keeling) Island, have been described by Hoernle et al. (2011).

Although there are numerous tectonic models for the evolution of the Northeast Indian Ocean and the western coast of Australia (Veevers, 1971; Heirtzler et al., 1978; Müller et al., 1998; Heine and Müller, 2005; Müller et al., 2008; Hall, 2012), none have integrated age constraints from volcanism in the area, with paleomagnetic constraints on the volcanic suite, to examine the causative mechanism of volcanism. In this paper, we present high resolution $^{40}Ar/^{39}Ar$ radioisotopic dating results to determine the ages and duration of volcanism on the island. In addition, this study presents a paleomagnetic analysis on the Christmas Island volcanics to determine the magnetic polarity of the lava flows, and compare these magnetic polarities with the geomagnetic timescale to constrain the reconstructed position of Christmas Island.

1.2. Geology of Christmas Island

Christmas Island is a roughly "T" shaped (Fig. 2) limestone capped island that lies 400 km south of the Java Trench and roughly 3000 km northwest of Perth. Its present position is 105° E, 10.5° S and is 360 m above sea-level, rising 5700–5900 m from the bottom of sea floor. Christmas Island has been previously surveyed essentially for exploitation of its groundwater aquifers (Polak, 1976; Pettifer and Polak, 1979) and phosphate mining operations (Doak et al., 1965; Trueman, 1965). Large parts of the island are now a world heritage site due to its unique fauna including the annual red crab migration (O'Dowd and Lake, 1989; O'Dowd and Lake, 1991).

The island is believed to have been formed initially in the Late Cretaceous as an undersea volcano (Grimes, 2001). It now sits on the bulge of the Australian plate subducting beneath the Sunda and Java Trenches which has led to its re-emergence above the sea-level (Grimes, 2001). Field studies of volcanic rocks at Sydney's Dale and Waterfall Spring suggest that the island was volcanically active in the Eocene but this had ceased by the Miocene (Barrie, 1967). Pettifer and Polak (1979) by visual examination of volcanic rocks and soils suggested, in their unpublished report, that the main phase of volcanism ceased in the Early Tertiary and the island became volcanically extinct in the Pliocene. There is no evidence of any volcanic activity in the Oligocene on the island and it has been suggested as a period of erosion (Barrie, 1967). The rise of the island due to the island riding the flexural bulge on the subducting plate is estimated to be around 140 mm per thousand years (Woodroffe, 1988) and has been subjected to sea-level fluctuation in the recent past (Grimes, 2001).

Barrie (1967) has reported the occurrence of a phosphatised volcanic material near the highlands on the western part of the island, more specifically, close to the Murray Hill region. In addition, it has been suggested that the volcanic outcrops at Flying Fish Cove and Dolly Beach are a result of faulting or wave cutting and are pre Miocene to early Miocene in age (Barrie, 1967). Baxter and Weeks (1984) have reported the volcanism in Pliocene was accompanied by faulting near the Murray Hill region. Some of the sub-aerial deposits fill limestone depressions, which suggests that the island was partly emergent in Early Pliocene (Grimes, 2001). Using geophysical investigations, Polak (1976) suggests the presence of rift zones to be responsible for a three pronged shape of the island. These rift zones partly cut deep erosion channels before the island emerged (Polak, 1976).

Works by Andrews (1900) and Trueman (1965) have documented three stages of volcanic eruptions with interbedded sequences of Tertiary limestone and phosphate rich soils. These interbeds have more specifically been exposed in the north near Flying Fish Cove and in the east close to Dolly Beach. The Late Cretaceous volcanic core of the island is buried, though earlier work has cited some exposures in the bottom of some limestone caves that are now being used for groundwater exploitation purposes, and are now inaccessible (Grimes, 2001). A Late Cretaceous phase of volcanism has been inferred from stratigraphic and faunal correlations (Grimes, 2001), and from flexural plate modelling (Taneja and O'Neill, in-review). It is overlain by basalts, basanites and trachytes of Middle to Late Eocene with limestones and dolomites (Table 1). The Pliocene phase of volcanism at the island occurs as minor volcanic dykes and tuffs that are overlain by phosphate rich soils.

2. Methodology and analytical techniques

2.1. Sample collection

Samples were collected during the field trip to the island in November, 2010. The rock cores were obtained using a water-cooled two-stroke powered motor drill. Samples were drilled in the field and oriented using a magnetic and a sun compass as described by Butler (1992) and McElhinny and McFadden (2000). Each individual core measured 3 to 5 cm in length with a diameter of 2.5 cm. Of the total sites sampled, three were on the east coast (Waterfall Spring, Ethel Beach and the Dolly Beach), two on the west coast (Winifred Beach and Sydney's Dale) and one site in north (Flying Fish Cove) as shown in Figs. 2 and 3. The three east coast sites and one western site (Winifred Beach) were used for paleomagnetism. All of the above mentioned sites were drilled except the Sydney's Dale due to its protected status under the Ramsay Convention on Wetlands. These four sites and additional hand samples from Sydney's Dale and Flying Fish Cover have been dated using $^{40}Ar/^{39}Ar$ geochronology. Dolly Beach on the east coast contains a vertically dipping basaltic dyke protruding through the shore, with a

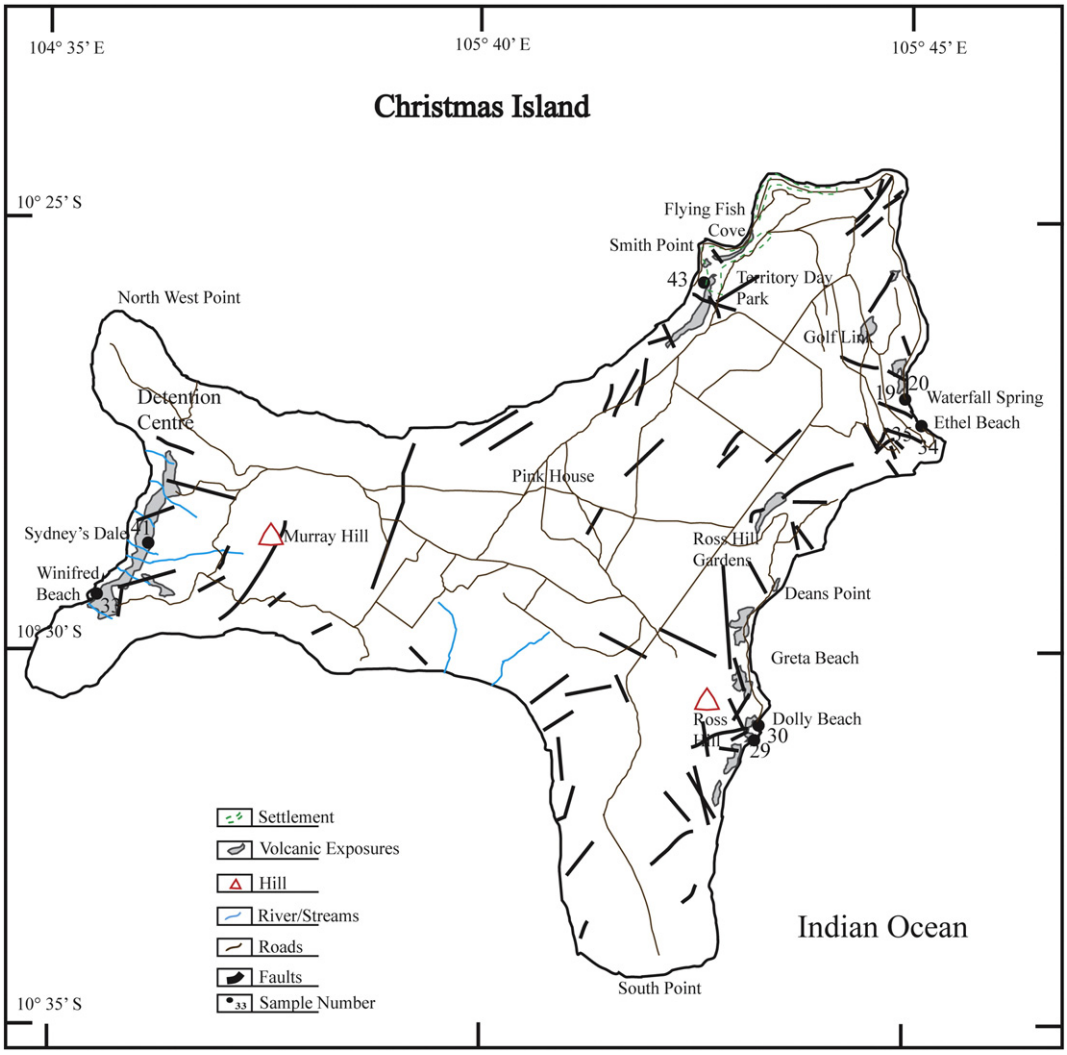


Fig. 2. A map of Christmas Island showing the location of volcanic sites sampled for geochronology and paleomagnetism. Volcanic exposures and road network re-drawn from maps provided by Christmas Island National Park. Settlement area modified from Christmas Island tourism maps.
Structural details are redrawn from Barrie (1967)

strike of ~45°, containing phenocrysts of olivine. The sub-horizontal sills at the base of the cliffs at Ethel Beach and Waterfall Spring are vesicular basalts. Winifred Beach and Sydney's Dale, the sampling sites on the west coast of the Island, outcrop inland and occur under a thick canopy of forest. Waterfall Spring and Ethel Beach are small outcrops spread across 3–5 m^2. Furthermore, the sampling sites on the east coast are located at the shore as compared to those on the west coast, which are at a higher elevation. Since most of the island is a dense tropical forest under the management of Parks Australia and a phosphate mining hub, access to certain parts of the island was limited or not possible, affecting sampling in some areas.

Table 1
Main geological units at the Christmas Islands, Northeast Indian Ocean, modified from Grimes, 2001.

Time period	Geological description
Quaternary	Phosphate deposits.
Pliocene (3–5 Ma)	Upper volcanic series, the youngest volcanism.
Late Oligocene–Mid Miocene	Upper carbonate series.
Eocene (35–40 Ma)	The lower volcanic series, second episode of volcanism.
Eocene	Lower carbonate series.
Late Cretaceous–Early Tertiary volcanic	The core volcanic part, mostly submerged and the oldest.

2.2. Geochronological techniques

Six samples were selected from Christmas Island for $^{40}Ar/^{39}Ar$ dating and were separated into unaltered, 600–800 μm-size groundmass. Groundmass particles were carefully hand-picked under a binocular microscope. Selected groundmass material (consisting of homogenous unoxidised pyroxene and plagioclase) was further leached in diluted HF for 1 min and then thoroughly rinsed with distilled water in an ultrasonic cleaner. Separates of plagioclase crystals have been shown to provide more accurate results than groundmass, as it facilitates the optical selection of the freshest grains (e.g., Hofmann et al., 2000; Jourdan et al., 2007; Verati and Jourdan, 2013), but their small size in our samples prevented us from using this approach.

Samples were loaded into six large wells of 1.9 cm diameter and 0.3 cm depth aluminium disc. These wells were bracketed by small wells that included Fish Canyon sanidine (FCs) used as a neutron fluence monitor for which an age of 28.305 ± 0.036 Ma (1σ) was adopted (Renne et al., 2010). The discs were Cd-shielded (to minimize undesirable nuclear interference reactions) and irradiated for 2 h in the Hamilton McMaster University nuclear reactor (Canada)



Fig. 3. Outcrop photographs during sample collection at Christmas Island, A, Waterfall Spring (XM-19C and XM-20); B, Dolly Beach (XM-29 and XM-30B); C, Ethel Beach (XM-34 and XM-35); D, Sydney's Dale (XM-40).

in position 5C. The mean J-values computed from standard grains within the small pits range from 0.00068800 ± 0.00000220 to 0.00071000 ± 0.00000099 determined as the average and standard deviation of J-values of the small wells for each irradiation disc. Mass discrimination was monitored using an automated air pipette and provided a range from 1.004492 ± 0.29 to 1.006127 ± 0.38 (%1σ) per dalton (atomic mass unit) relative to an air ratio of 298.56 ± 0.31 (Lee et al., 2006). The correction factors for interfering isotopes were $(^{39}Ar/^{37}Ar)_{Ca} = 7.30 \times 10^{-4}$ (±11%), $(^{36}Ar/^{37}Ar)_{Ca} = 2.82 \times 10^{-4}$ (±1%) and $(^{40}Ar/^{39}Ar)_{K} = 6.76 \times 10^{-4}$ (±32%) (e.g., Jourdan and Renne, 2007).

The $^{40}Ar/^{39}Ar$ analyses were performed at the Western Australian Argon Isotope Facility at Curtin University. The samples were wrapped in degassed niobium foil packages and step-heated using a 110 W Spectron Laser Systems, with a continuous Nd-YAG (IR; 1064 nm) laser rastered over the sample for 1 min to ensure a homogenously distributed temperature. The gas was purified in a stainless steel extraction line using two SAES AP10 and one GP50 getter and a liquid nitrogen condensation trap. Argon isotopes were measured in static mode using a MAP 215-50 mass spectrometer (resolution of ~450; sensitivity of 4×10^{-14} mol/V) with a Balzers SEV 217 electron multiplier mostly using 9 to 10 cycles of peak-hopping. The data acquisition was performed with the Argus programme written by M.O. McWilliams and ran under a LabView environment. The raw data was processed using the ArArCALC software (Koppers, 2002) and the ages have been calculated using the decay constants recommended by Renne et al. (2010). Blanks were monitored every 3 to 4 steps and typical ^{40}Ar blanks range from 1×10^{-16} to 2×10^{-16} mol. Ar isotopic data corrected for blank, mass discrimination and radioactive decay are given as Supplementary material. Individual errors (given in the Supplementary material) are given at the 1σ and 2σ levels, while Age and K/Ca (Fig. 5) are given at 2σ level. Our criteria for the determination of plateau are as follows: plateaus must include at least 70% of ^{39}Ar. The plateau should be distributed over a minimum of 3 consecutive steps agreeing at 95% confidence level and satisfying a probability of fit (P) of at least 0.05. Plateau ages (Fig. 5) are given at the 2σ level and are calculated using the mean of all the plateau steps, each weighted by the inverse variance of their individual analytical error. Mini-plateaus are defined similarly except that they include between 50% and 70% of ^{39}Ar. Inverse isochrons include the maximum number of steps with a probability of fit ≥0.05. All sources of uncertainties are included in the calculation.

2.3. Paleomagnetic laboratory techniques

The drilled cores were cut into 2.2 cm individual core samples and used for paleomagnetic analysis. A select set of samples were demagnetised using an alternating-field progressive demagnetiser (Model 2G600) and another separate set of samples were thermally demagnetised using MMTD80 programmable thermal demagnetiser at CSIRO, North Ryde. The oven within the demagnetiser is surrounded by 4 layers of mu-metal shielding with a residual field of less than 10 nT. It is capable of heating up to 80 one inch samples automatically. Remanent intensities were measured using 2G Enterprise 3-axis cryogenic magnetometer (Model 755). Alternating field demagnetisation was carried out in fourteen demagnetisation steps mostly with increments of 5–10 mT up to maximum of 70 mT. Selected samples underwent thermal demagnetisation from 200 °C in increments of 50 °C up to 500 °C, thereafter in increments of 5–20 °C up to 570 °C.

Magnetic susceptibilities were measured using a Sapphire Instruments SI2B susceptibility instrument.

The demagnetisation steps with their inclinations and declinations were plotted on orthogonal vector diagrams. Magnetic remanence locked within the sample consists of a primary magnetic component, and often associated with it is later secondary magnetisation. This auxiliary magnetic character is acquired, in geological time, due to secondary processes like viscous acquisition or effects of stray fields from lightning. Metamorphism of the rock body, or alteration due to chemical reactions, can both leave secondary or later stage magnetic imprints. Interpretation of demagnetisation was done by preparing visually identifiable univectorial decay vectors on orthogonal vector plots, called orthogonal diagrams (Zijderveld, 1967), to determine the polarity and progressive demagnetisation. This helps in identifying the primary magnetic component, also called Characteristic Remanent Magnetisation (ChRM), and differentiating it from secondary components. Principal component analysis (PCA) was used in calculating mean directions and mean intensity of ChRM (Kirschvink, 1980). Fisherian analysis is used for calculation of mean inclination (I_m) and declination (D_m) for each site as shown in Tables 4 and 5, using the software Paleomagnetic Tools (PMAGTOOL) V 4.2a, and an in-house software developed by Phil Schmidt, CSIRO. Mean inclination, of a sample can be used to calculate the paleolatitude by a simple relation,

$$\tan I_m = 2\tan\lambda \quad (1)$$

where, λ, is the latitude of the paleosite.

3. Results

3.1. Petrology

Only fresh samples from Christmas Island were selected for geochronology, and in-situ sites were drilled for paleomagnetic analysis. The Dolly Beach site on the east cost of the island consists of a vertical basaltic dyke emerging from the coast. Petrographically (Fig. 4), it contains a fine-grained groundmass (~70 vol.%) consisting of microliths of plagioclase (50 μm to 200 μm), glass, and opaque minerals. Phenocrysts of olivine (25 vol.%) and clinopyroxene (5 vol.%) form the remaining 30% of the sample. Phenocrysts of olivine vary from <50 μm to several cm and are euhedral to subhedral. Magnetite inclusions (~10 μm) occur in pyroxenes within the groundmass, and occasionally surrounding grain boundaries of olivine. Waterfall Spring and Ethel Beach are very similar petrographically, with a fine-grained groundmass (plagioclase, groundmass and opaque oxides) that comprises ~80 vol.% of the sample. Plagioclase occurs as elongate euhedral to subhedral needles that are randomly oriented and range from 50 μm to over 100 μm. Phenocrysts of olivine (15 vol.%) are fractured (affected by serpentinisation), and are euhedral to subhedral (100 μm to 400 μm), while the remaining 5 vol.% comprises clinopyroxenes (50 μm–200 μm). Needle shaped magnetite (50 μm) are present within chromite. Flying Fish Cove, on the north side of the island, consists of phenocrysts of olivine (150 μm to several cm), and minor clinopyroxene comprising total 30 vol.% of the sample, while the remaining part is fine-grained groundmass. Plagioclase averages 100 μm in size, and forms

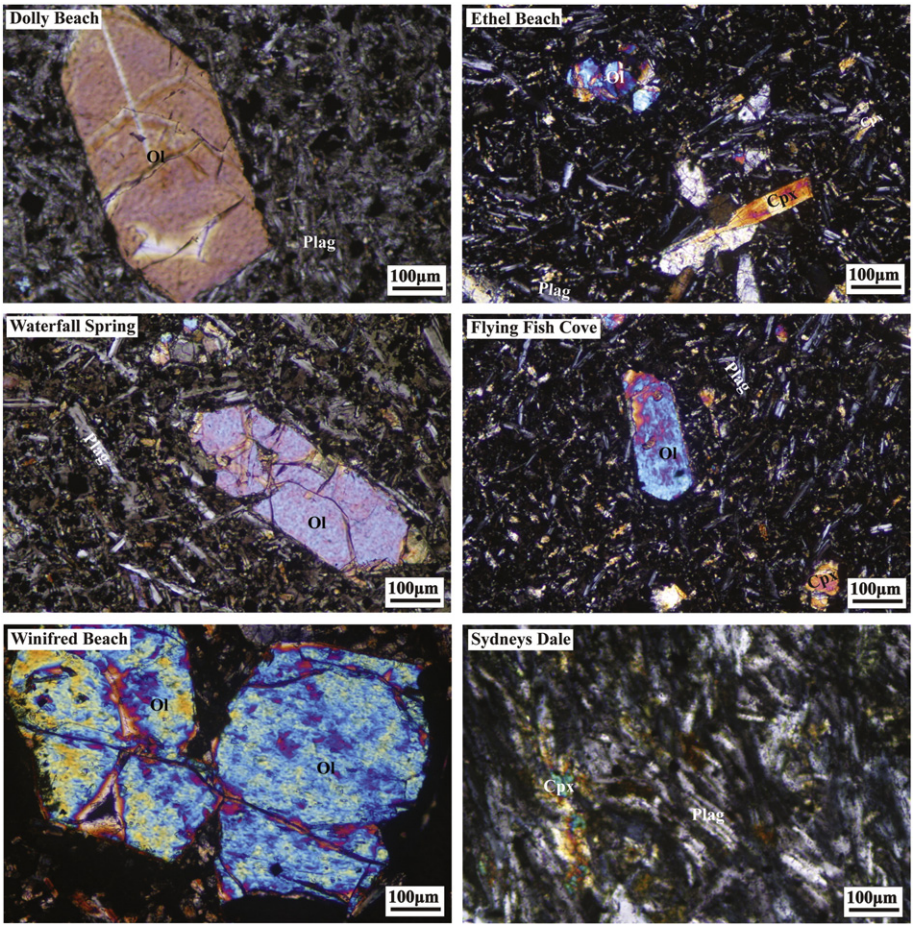


Fig. 4. Petrological examination of volcanic rocks from Christmas Island. Primary mineralogy consists of phenocrysts of olivine and clinopyroxene, in a fine grained matrix comprising of plagioclase, glass, and opaque oxides. Modal abundance of groundmass in volcanic rocks from Christmas Island varies between 70 vol.% and 80 vol.%. Phenocrysts of olivine are euhedral to subhedral (~50 μm to several mm), while plagioclase occurs as needle like ranging from 50 μm to 300 μm. All photos taken in crossed polars.

80 vol.% of the groundmass volume, while 20 vol.% of the groundmass is glass and opaques, both occurring in equal proportions. Some sections of the sample have evidence of alteration that appears to have affected grain boundaries of olivine. Winifred Beach samples exhibit a fine-grained groundmass of clinopyroxene (25 vol.%, 25 μm to 200 μm) that are euhedral to subhedral in shape. Plagioclase laths (60 vol.%, 25 μm to 150 μm) form the majority of the groundmass, together with minor opaques and glass (15 vol.%). Olivine is the primary phenocryst observed (~70 vol.% of the total phenocrysts), with minor clinopyroxene (~30 vol.%). Randomly oriented needle-like magnetites appear within the chromite. Sydney's Dale exhibits a characteristic trachyte texture of flow-oriented mineral grains. The samples are very fine grained, making it difficult to identify minerals, but the sample predominantly consists of oriented plagioclase (50 μm to 200 μm), with opaque and glass and rare clinopyroxene. Plagioclase appears to have undergone sericitic alteration, and glass and earlier mafic phases to chlorite.

3.2. $^{40}Ar/^{39}Ar$ dating

The step wise laser heating experiment for sample from Dolly Beach (XM-30B) gave a plateau age of 43.55 ± 0.44 Ma (MSWD = 1.4; P = 0.14; Fig. 5) including 99% of ^{39}Ar released. The K/Ca (Fig. 5) (derived from $^{39}Ar_K$ and $^{37}Ar_{Ca}$) spectra show a slight variation typically observed for basalt samples (McDougall and Harrison, 1999). The inverse isochron (Fig. 5) gives an age of 43.46 ± 0.66 Ma (MSWD = 1.4; P = 0.14) and with a $^{40}Ar/^{36}Ar$ intercept of 300.8 ± 2.5, indistinguishable from the atmospheric ratio of 298.6 ± 0.3 (Lee et al., 2006) adopted in this study. This indicates that no excess argon is present in this sample, and hence we take the plateau age as indicating the time since the eruption of the magma.

Ethel Beach (XM-34) shows a two-level age spectrum suggesting that alteration perturbed this sample to some extent. A mini-plateau (minimum) age of ≥37.10 ± 0.66 Ma (MSWD = 0.86; P = 0.56) including 63% of the ^{39}Ar released, suggests that the effects of alteration were not extensive (Fig. 5). The shape of the K/Ca spectrum matches the shape of the age spectrum and shows a decrease of the K/Ca ratio towards high temperature steps (Fig. 5). The inverse isochron yielded an age of 36.3 ± 2.1 Ma (MSWD = 0.9; P = 0.53) and a $^{40}Ar/^{36}Ar$ trapped ratio of 316 ± 20, indistinguishable from the atmospheric ratio. We conservatively interpret the plateau age as being a minimum age, albeit probably close to the eruption age of this sample.

Sample XM-41 at Sydney's Dale on the west coast did not yield a plateau age as the age spectrum shows a barely visible, but nevertheless individual, step age variation. However, it did yield a well-behaved inverse isochron age of 41.75 ± 0.30 Ma (MSWD = 0.85; P = 0.63) including 92% of cumulative ^{39}Ar and an intercept $^{40}Ar/^{36}Ar$ trapped value of 306.1 ± 3.4 (Fig. 5). The $^{40}Ar/^{36}Ar$ value suggests that some excess $^{40}Ar^*$ is present in the sample, but is fully accounted for in the inverse isochron age calculation (e.g. Sharp and Renne, 2005; Jourdan et al., 2012). As an exercise, we calculated a plateau age of 41.75 ± 0.18 Ma (MSWD = 0.75; P = 0.75) using a $^{40}Ar/^{36}Ar$ value of 306.1 instead of 298.6 (Fig. 5) but we retain the inverse isochron age for the discussion.

The results for the sample from Flying Fish Cove (XM-43) near the north shore of the island, however, shows a more disturbed age spectrum (Fig. 5). Such a spectrum with decreasing apparent ages at high temperature is characteristic of an alteration overprint, where the effects of alteration (usually neo-crystalized K-rich phases) on apparent step ages are more apparent on K-poor domains compare to K-rich domains. A contribution from alteration cannot be excluded from the low temperature steps and probably lowers the age of the mini-plateau. It only provides qualitative constraints for the age of this sample, with a minimum age of ≥38.7 ± 0.5 Ma (MSWD = 0.48; P = 0.87) as given by a mini-plateau including 56% of the ^{39}Ar released. The K/Ca spectrum show little compositional zoning in the sample (Fig. 5). The inverse isochron shows only little spread, but the trapped ratio (319 ± 71), although imprecise, seems to suggest that no excess $^{40}Ar^*$ is present.

Waterfall Spring (XM-19) yielded a plateau over 100% of the ^{39}Ar released and gave an age of 37.75 ± 0.77 Ma (MSWD = 0.65; P = 0.75, Table 2). The K/Ca spectrum suggests little zoning in this sample (Fig. 5). The inverse isochron has a small spreading factor preventing an estimate of the proper age and intercept value (Fig. 5).

Winifred Beach (XM-33) on the other hand is the youngest sample, and gave a plateau age of 4.32 ± 0.17 Ma (MSWD = 1.56; P = 0.15) including 79% of cumulative ^{39}Ar (Fig. 5). The age spectrum shows that the step ages get slightly younger over the last 21% of the age spectrum, possibly due to minor alteration. The breadth of the plateau suggest that it is not affected by any secondary process, and we interpret the $^{40}Ar/^{39}Ar$ plateau age as indicating the eruption age of this sample. The inverse isochron yielded an imprecise intercept $^{40}Ar/^{36}Ar$ ratio of 309 ± 49, indistinguishable of atmospheric composition. The K/Ca ratio suggests that very little Ca was present in this sample and there is more glass and groundmass.

3.3. Paleomagnetic analysis

The Koenigsberger ratio (Q) is the ratio of the remanent magnetization to the induced magnetization (product of susceptibility and the Earth's magnetic field strength). It is an effective parameter to provide an indication of the significance of remanence magnetism. A large Q implies that the magnetic material will tend to retain significant remanent magnetization and hence is a strong magnetic recorder. Nakanishi and Gee (1995) calculated the Q values for north-western Pacific guyots and concluded that these are reliable enough for paleolatitude determination. The Koenigsberger ratio (Q ratio) mathematically, is defined as:

$$Q = \frac{NRM}{k * H} \quad (2)$$

where, NRM, is the remanent magnetism measured before AF demagnetisation; k is the susceptibility; H is the total magnetic field at Christmas Island. Both remanent and field intensities are measured in units of A/m.

Relatively high Q values for most of the samples as shown in Table 3 make them suitable for paleomagnetic analyses. The orthogonal projections displaying magnetic vector components are the most common tool to determine and analyse paleomagnetism properties by selecting vector end points that decay univectorially towards the origin as shown in Figs. 6 and 7. The Ethel Beach (XM-35) and Waterfall Spring (XM-20) sampling sites are located on the east coast, and are ~300 m apart. The assumed primary ChRM for both the sites is resolvable after 10–15 mT demagnetisation steps. Both sites have a mean declination that is northwards with a steep inclination of −57.2° and −65.5° (directed upwards) respectively. The two sites being in close proximity of each other and their similar geochronology age imply that both of the sites have recorded the same event of magnetism and magma solidification.

Dolly Beach (XM-29) on the east coast has a steeper inclination of 77.6° that points down, opposite to Waterfall Spring and Ethel Beach. Stereographic projection (Fig. 6) shows a declination centred near the south. Winifred (XM-33), on the west coast is the youngest at ~4.3 Ma and has a primary magnetism pointing east-southeast in the down direction. The inclination points in the same direction as the oldest site (XM-29); it is, however, opposite to Waterfall Spring and Ethel Beach (Fig. 6). Samples from Winifred Beach were demagnetised thermally in addition to AF-demagnetisation (Fig. 7). This proved to be more effective in resolving the characteristic component of up to 500 °C. Beyond this temperature erratic behaviour was observed for thermally demagnetised specimens indicating that the samples had been demagnetised.

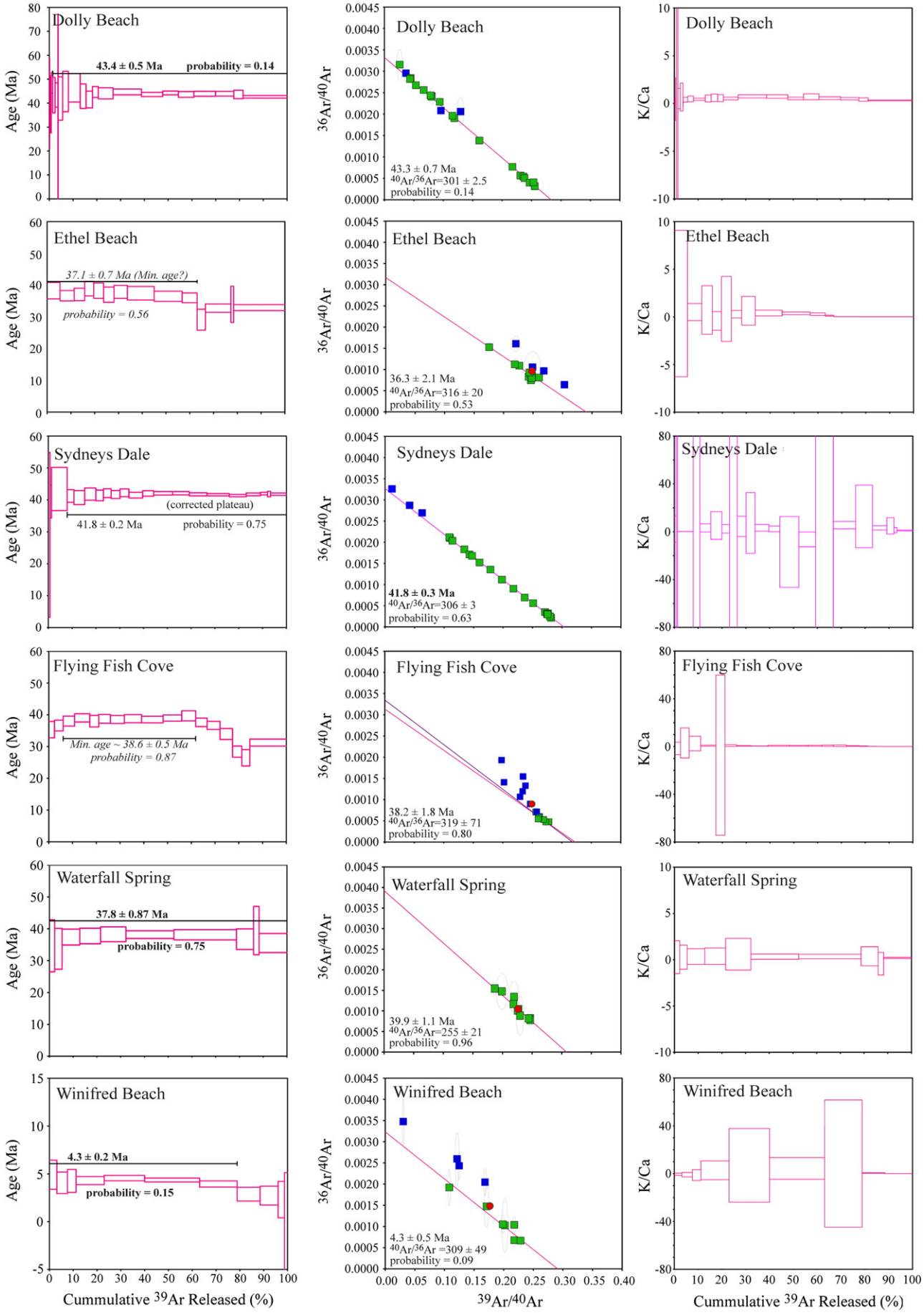

Dolly Beach
43.4 ± 0.5 Ma
probability = 0.14
Age (Ma)
Dolly Beach
43.3 ± 0.7 Ma
40Ar/36Ar=301 ± 2.5
probability = 0.14
36Ar/40Ar
Dolly Beach
K/Ca
Ethel Beach
37.1 ± 0.7 Ma (Min. age?)
probability = 0.56
Age (Ma)
Ethel Beach
36.3 ± 2.1 Ma
40Ar/36Ar=316 ± 20
probability = 0.53
36Ar/40Ar
Ethel Beach
K/Ca
Sydneys Dale
(corrected plateau)
41.8 ± 0.2 Ma
probability = 0.75
Age (Ma)
Sydneys Dale
41.8 ± 0.3 Ma
40Ar/36Ar=306 ± 3
probability = 0.63
36Ar/40Ar
Sydneys Dale
K/Ca
Flying Fish Cove
Min. age ~ 38.6 ± 0.5 Ma
probability = 0.87
Age (Ma)
Flying Fish Cove
38.2 ± 1.8 Ma
40Ar/36Ar=319 ± 71
probability = 0.80
36Ar/40Ar
Flying Fish Cove
K/Ca
Waterfall Spring
37.8 ± 0.87 Ma
probability = 0.75
Age (Ma)
Waterfall Spring
39.9 ± 1.1 Ma
40Ar/36Ar=255 ± 21
probability = 0.96
36Ar/40Ar
Waterfall Spring
K/Ca
Winifred Beach
4.3 ± 0.2 Ma
probability = 0.15
Age (Ma)
Winifred Beach
4.3 ± 0.5 Ma
40Ar/36Ar =309 ± 49
probability = 0.09
36Ar/40Ar
Winifred Beach
K/Ca
Cummulative 39Ar Released (%)
39Ar/40Ar
Cummulative 39Ar Released (%)

Table 2
$^{40}Ar/^{39}Ar$ analysis for six samples from Christmas Island; p, probability of fit (P) of at least 0.05; MSWD, mean square weighted deviation; Age in bold are the age adopted in this study. Ages underlined and accompanied with the sign ≥ are minimum ages.

Sample no.	Location	Plateau age				$^{39}Ar(k)$ released (%)	Inverse isochron			$^{40}Ar/^{36}Ar$	
		Age	2σ	MSWD	p		Age	2σ	MSWD	Intercept	2σ
XM-34	Ethel Beach	≥37.1	0.66	0.86	0.56	63	36.29	2.08	0.88	316	20
XM-41	Sydney's Dale	41.75	0.18[a]	0.75	0.75	92	**41.75**	**0.30**	0.85	306	3.0
XM-30B	Dolly Beach	**43.55**	**0.44**	1.37	0.14	99	43.46	0.66	1.38	301	2.5
XM-19C	Waterfall Spring	**37.75**	**0.77**	0.65	0.75	100	39.97	1.11	0.3	255	21
XM-43	Flying Fish Cove	≥38.7	0.5	0.48	0.87	56	38.20	1.83	0.54	319	71
XM-33D	Winifred Beach	**4.32**	**0.17**	1.56	0.15	79	4.27	0.46	1.85	309	49

[a] Calculated using the measured $^{40}Ar/^{36}Ar$ intercept ratio.

Furthermore, two sites (Dolly Beach and Winifred) have positive inclinations. Assuming a southern hemisphere origin of these volcanics, this denotes that the Earth's magnetic polarity reversed while magnetism was locked within these sites.

These are very large errors associated with some of the site VGP's (Table 4) in the paleomagnetic analysis and hence appear doubtful if these can be used effectively to find paleoposition. Therefore, a combined position was obtained for the Eocene volcanism at the Christmas Island (Table 5) by combining all the Eocene dates. In addition, the Fisher mean of the individual fisher means (means of the three Eocene VGP sites) was calculated to constrain the errors (Table 5).

4. Discussion

4.1. Duration of volcanism

$^{40}Ar/^{39}Ar$ geochronology shows multiple stages of volcanism at Christmas Island. Two suites of volcanism at the island have been dated and identified, an older Lower Volcanic Suite (LVS), and a younger Upper Volcanic Suite (UVS) (Fig. 8). The oldest ages within the LVS are found at Dolly Beach on the east coast and Sydney's Dale on the west coast. We believe that this volcanism initiated in the Middle Eocene between 42 and 43 Ma. Further on, Late Eocene (~38–39 Ma) volcanism is evident at Waterfall Spring and Ethel Beach on the east coast, and Flying Fish Cove in the north. Hoernle et al. (2011) conducted $^{40}Ar/^{39}Ar$ geochronology analyses that yielded a lowest age of 37.0 ± 0.6 Ma, and an oldest age of 43.6 ± 0.4 Ma. Hoernle et al. (2011) also obtained intermediate ages of 39.6 ± 0.4 Ma, 40.2 ± 0.2 Ma, 41.1 ± 0.1 Ma and 42.6 ± 0.4 Ma. Combining ages from this study and those obtained by Hoernle et al. (2011), it appears that volcanism on the island was continuous from 37 Ma to 43 Ma.

The last and final episode of volcanism at the island started after a time gap of ~33 Myr in the Early Pliocene at 4.32 ± 0.17 Ma. Hoernle et al. (2011) yielded similar young ages in the range of 4.31 ± 0.14–4.52 ± 0.24 Ma. New high resolution radioisotopic dating of the basaltic rocks obtained from this study and those dated by Hoernle et al. (2011) confirms two dated episodes of volcanism at the island, and it strongly contradicts a middle phase in the Miocene as argued by Barrie (1967). A Late Cretaceous volcanic core of the island is, however, not exposed hence no dates are available for it. However, stratigraphic studies on interbedded limestones on the Island (Grimes, 2001), and recent plate flexural modelling (Taneja and O'Neill, in-review) have suggested a Late Cretaceous volcanic episode, which led to elastic loading of the young oceanic lithosphere around 75–88 Ma.

4.2. Petit spot low volume volcanism

Low volume "petit spot" volcanism formed away from the mid-oceanic ridge and subduction zones has been recently described off the north-western Pacific plate (Hirano et al., 2001, 2006). Hirano et al. (2008) calculated the volume of these petit spot monogenetic volcanoes and suggested that these are usually <1 km^3. These are further differentiated on the basis of acoustic reflectivities with the petit spot features having a higher reflectance than those compared to the older and much larger submarine volcanoes (Fujiwara et al., 2007).

Oceanic lithosphere behaves as an elastic sheet when loaded, for stresses below its intrinsic yield stress (Bodine et al., 1981; Watts, 2001). Hirano et al. (2001, 2006) while discussing the petit spot intraplate volcanism at Hokkaido Rise, in the northwest Pacific Ocean suggested that these were formed due to fracturing within the flexed outer rise of the lithospheric bulge. The flexure induced by bending of the plate as it approaches a subduction zone causes redistribution of internal stresses leading to fracturing of the lithosphere that may cause decompression melting in subjacent asthenospheric mantle, potentially allowing the ascent of small volumes of melt in nascent cracks (Hirano et al., 2001). The Indo-Australian plate in the region below the Java and Sunda Trenches is undergoing subduction and demonstrates flexure on the outer rise of the subducting plate (Levitt and Sandwell, 1995; Smith and Sandwell, 1997). A cross section of such a flexure produced along a line perpendicular to the trench passing through the island is shown in Fig. 9. The flexure produced by the load of Christmas Island is overprinted on the crest of the flexed lithosphere as it subducts at the trench. A similar local flexure is produced by Golden Bo'Sunbird, a seamount, south west of Christmas Island as shown in Fig. 9.

Magma flow rising through these cracks reaches the surface in the form of petit spot volcanism. The younger episode of volcanism on Christmas Island may share some characteristics with these low-volume eruptions. Even though there are no constraints on the volume of volcanism at Christmas Island, as it is covered by thick sequences of limestone, the exposure of the Pliocene phase of volcanism at Winifred Beach on the west coast of the island, emplaced through the older volcanic core and limestone cliffs and terraces, bears similarity to this petit spot model due to the following reasons. Baxter and Weeks (1984) suggested that the volcanism in the region around the Murray Hill was accompanied by faulting that led to volcanic vents, fractures and dykes. Hoernle et al. (2011) dredged Pliocene volcanism around the flanks of Christmas Island and this study has sampled one location on the island. The limited occurrences of the younger phase of volcanism, far less than the older dated samples, and its association with faulting and emplacement as dykes suggest that this is a small phase.

Fig. 5. Age plateau (left), inverse isochron (centre), and K/Ca (right) plot for the six sampling sites at Christmas Island. Uncorrected sample from Sydney's Dale did not produce an age plateau, but an age was obtained from Inverse Isochron (highlighted in bold), and a corrected sample yielded an age which is shown above. Squares in green are those used for age calculations. Ages in bold are the ages adopted in this study. Ages in italics are minimum ages.

Table 3
Koenigsberger ratio (Q) for paleomagnetic samples from Christmas Island. The total magnetic field at Christmas Island is 46800 nT (H = 37.2 A/m). The high intensity for Winifred Beach represents minimum value measured and could be stronger, thus their Q values could be higher, while the Q values for other sites are well within the ranges measured by 2G and are precise.

Location	Sample	Susceptibility, SI	NRM (mA/m)	Q
Winifred Beach	XM-33A2	0.01145	17790	42.43
	XM-33B2	0.01042	18810	49.27
	XM-33B3	0.01058	17930	46.28
	XM-33C2	0.01142	16880	40.35
	XM-33E2	0.01107	17650	43.54
	XM-33A1	0.01173	17790	41.40
	XM-33B1	0.01050	16860	43.83
	XM-33C1	0.01147	17080	40.64
	XM-33D1	0.01058	18870	48.72
	XM-33E1	0.01108	18220	44.89
Waterfall Spring	XM-20C2	0.00130	1070	22.55
	XM-20A1	0.00131	1340	27.95
	XM-20C1	0.00119	1090	25.22
	XM-20D1	0.00227	2930	35.22
Dolly Beach	XM-29B2	0.00528	8280	42.85
	XM-29C2	0.00529	4280	22.09
	XM-29B1	0.00579	7050	33.28
	XM-29C1	0.00569	8210	39.40
	XM-29D1	0.01229	9690	21.54
Ethel Beach	XM-35A1	0.00316	8210	71.04
	XM-35B1	0.00256	4150	44.30

Additionally, if the Pliocene phase of volcanism is indeed produced due to flexure, then the onset of volcanism should correlate with the onset of flexure, which can be determined from present observations of the forebulge morphology, using plate reconstructions and past plate velocities. To determine the paleoposition of the erupted younger volcanics, we backtracked the island from its present position at 10.5° S using a plate motion rate of 7 cm/yr (Tregoning et al., 1994). Tracing the island back to its position at 4.32 ± 0.17 Ma, we calculate its reconstructed position 302 ± 11.9 km (12.70°–12.85° S) southwest of its present day location as shown in Fig. 9. Two additional small volcanic features are present between the point of lithospheric cracking, and Golden Bo'Sunbird seamount. These two features might represent petit spot volcanic features documented by Hirano et al. (2001) for which additional dredging expeditions should be carried out. This, however, is beyond the scope of the current study.

Hirano et al. (2006) report petit spot volcanism could represent a new intraplate type volcanic setting. They, along with Buchs et al. (2013), are of the opinion that petit spot volcanism has been inadequately sampled, compared to other known volcanic islands in the Pacific and elsewhere, and that this volcanic setting could be more common than previously thought, and it also presents a tremendous scope for further geophysical, geochemical and geodynamic investigations.

4.3. Magnetic polarity and paleoposition

The Eocene epoch has been marked by magnetic reversals and paleomagnetism analysis can uncover changes in magnetic polarity locked within a rock. This study matched magnetic polarity and geochronological ages with Cande and Kent's (1995) and Wei's (1995) Geomagnetic Polarity Time Scale (GPTS) in a comparison with Earth's paleomagnetic field record. Waterfall Spring and Ethel Beach, with ages of 37.8 ± 0.8 Ma and 37.1 ± 0.7 Ma respectively, have normal polarity (present day). Dolly Beach, with an age of 43.4 ± 0.5 Ma, has a positive inclination (Table 5), implying a reverse paleomagnetic field. Dolly Beach falls within polarity chron C20n–C20r as summarised in Table 6. Winifred Beach, with an age of 4.3 ± 0.2 Ma, has a reverse polarity as evident from positive inclination in stereographic projection in Fig. 7 and Table 6.

Paleomagnetic investigations for Dolly Beach and Waterfall Spring have very steep inclinations, and this indicates a high paleolatitude for these sites. Polak (1976) found similar high inclinations for two of the three sites where measurements were taken. An inclination of −65.5° has been reported for Waterfall Beach and 67.9° for Dolly Beach — similar and comparable to our analysis. The third site on the west coast near Winifred Beach has a low inclination of 16.1°. The magnetic polarity of both data is similar. Polak (1976) has, however, not given details regarding the paleomagnetic analysis of these samples. The results are based on just one sample per site and it is not clear if drilled, or in-situ samples were used for these calculations — lumps and chips have been mentioned while describing some rock properties.

Paleomagnetic analyses of the Ethel Beach, Waterfall Spring and Dolly Beach have given a VGP latitudes of $-37.8^{\circ}{}^{+19.4^{\circ}}_{-34.1^{\circ}}$; $-47.6^{\circ}{}^{+7.1^{\circ}}_{-8.3^{\circ}}$ and $-66.2^{\circ}{}^{+19.9^{\circ}}_{-22.3^{\circ}}$ respectively. There are large error associated with individual site VGP's, hence the average of the Eocene volcanism was calculated as shown in Table 5. By doing this, multiple flows of lava are generally included within the combined data and are more effective in averaging any secular variation experienced while the basaltic body cooled. The pole calculated from combining the results can only be considered to be a virtual geomagnetic pole (VGP). The paleolatitude is therefore not tightly constrained and can only be taken as indicative. While we tried to maximise the sampling of volcanic flows a lack of suitable outcrops limits the statistical averaging of the VGPs. Same can be said for the Winifred Beach where a single flow sampled may record a transient field during a reversal. The combination of Fisher means of the 3 VGP sites (Table 5) gives a combined paleolatitude of $-49.7^{\circ}{}^{+19.0^{\circ}}_{-28.7^{\circ}}$ (with a full range of paleolatitude between −30.47° and −78.4°) This position is further south than the reconstructed position of the Island as using a moving hotspot model (O'Neill et al., 2005), which positions the island at 30° S at 45 Ma, and 28° S at 37 Ma. The position of the Island during most recent phase of volcanism during the Pliocene, at ~5 Ma, would be around 11.5 – 12° S.

The drilling operations were carried out using a portable hand drill and it is probable that an insufficient number of flows were sampled within the sites to reduce the errors. Similar issues were encountered by Nakanishi and Gee (1995) in their magnetic studies in the Northwestern Pacific Ocean Guyots, where they suffered from secular variations and subsequent differences in paleolatitudes. Dupont-Nivet et al. (2010), while calculating paleolatitude and the age of Indo-Asia collision suggested $k > 50$ and $n \geq 5$ as a sufficient dataset to represent time averaged geomagnetic variations. Montes-Lauar et al. (1995) adopted a methodology of using two samples per site to determine the mean, they admit poor statistical quality in the data but consider worth including these due to consistent directions.

Hence, instead of using the three Eocene sites, like above, the study combines the two Late Eocene (Waterfall Beach and Ethel Beach) sites, to get a slightly shallower paleolatitude of $-43.5^{\circ}{}^{+9.0^{\circ}}_{-11.2^{\circ}}$ (with a full range of paleolatitude between −34.4° and −54.7°). This is still quite different from the reconstructed position, but the errors induced are significantly smaller. It seems quite reasonable to combine the two sites because, firstly, they have similar ages (Table 2) and secondly, they are separated by 300 m and therefore, represent different lava flows of the same sequence, whereas Dolly Beach is 7–9 km south of the two sites. By combining these two events, we hope to have sampled at least two flows of the same event. And with an increased number of samples we hope to have constrained secular variations slightly better as compared to individual sites. Given the relationship between the two sites, it is possible that the recalculated paleolatitude might be a spot reading of the geomagnetic field, and should be considered indicative. Moreover, the reconstructed position is now closer to the position ascribed by reconstruction model of O'Neill et al. (2005) of ~30° S, when the error range is considered.

Seismic tomography models of Grand (2002) and Montelli et al. (2004) document the presence of a low velocity zone beneath the Eocene position of the Christmas Island as shown in Fig. 10. Grand's (2002) seismic tomography model has very clearly documented a seismic low velocity zone underneath the reconstructed position of

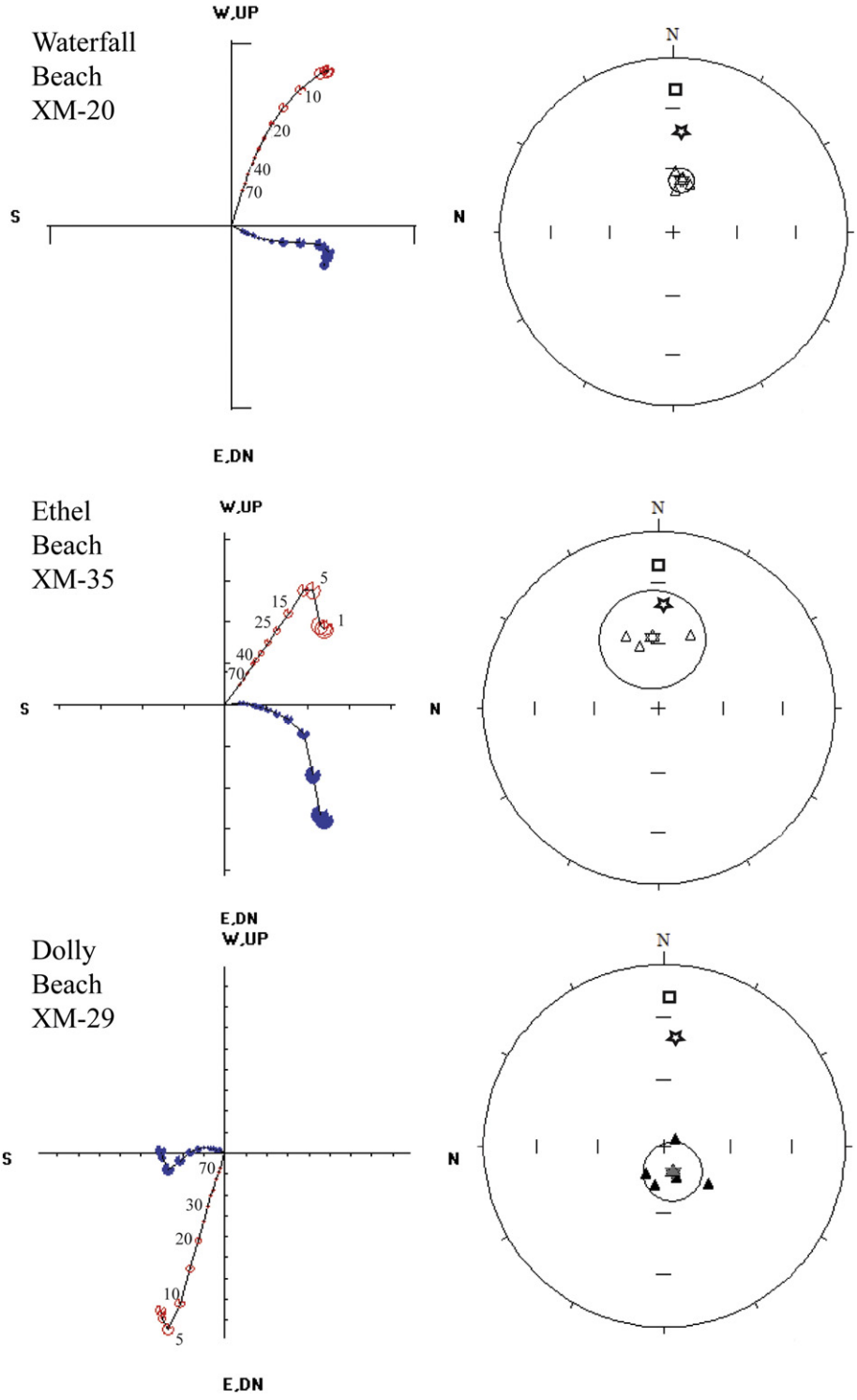


Fig. 6. Representative orthogonal vector plots (left) and site stereonet plots (right) illustrating demagnetisation. Blue (solid) data points are projected onto the horizontal plane, red (open) data points are projected onto vertical plane. First tick on orthogonal vector plot represents 1000 mA/m. Black star (within stereonet) is the present magnetic field, D = 1.5°, I = −39° and the Black square is the dipole field at Christmas Island.

Christmas Island in the Eocene down to at least 900 km, and potentially deeper down to 1600 km, beyond which the resolution is poor. Montelli et al. (2004) documented a plume in the East Indian Ocean using both P and S waves centred around 35° S and 100° E down to a depth of 1900 km below which they argue the resolution is poor. They also discussed a deep plume underneath the Cocos (Keeling) Island and a starting plume South of Java. Therefore the association of deep seated low velocity seismic anomalies around the reconstructed position of Christmas Island is striking (Montelli et al., 2004), and may be implicated in the re-emergence of volcanism at Christmas Island.

Oceanic islands are susceptible to tilting as they begin to ride the bulge produced due to subduction at the trench. Tilt associated with Christmas Island is between 0.23° and 1.27° and the Island presently sits right at the crest of the bulge (Fig. 9). The effect of tilt is more clearly seen at the Golden Bo'Sunbird seamount, which is riding the bulge, south-west of Christmas Island. After correcting the inclinations for

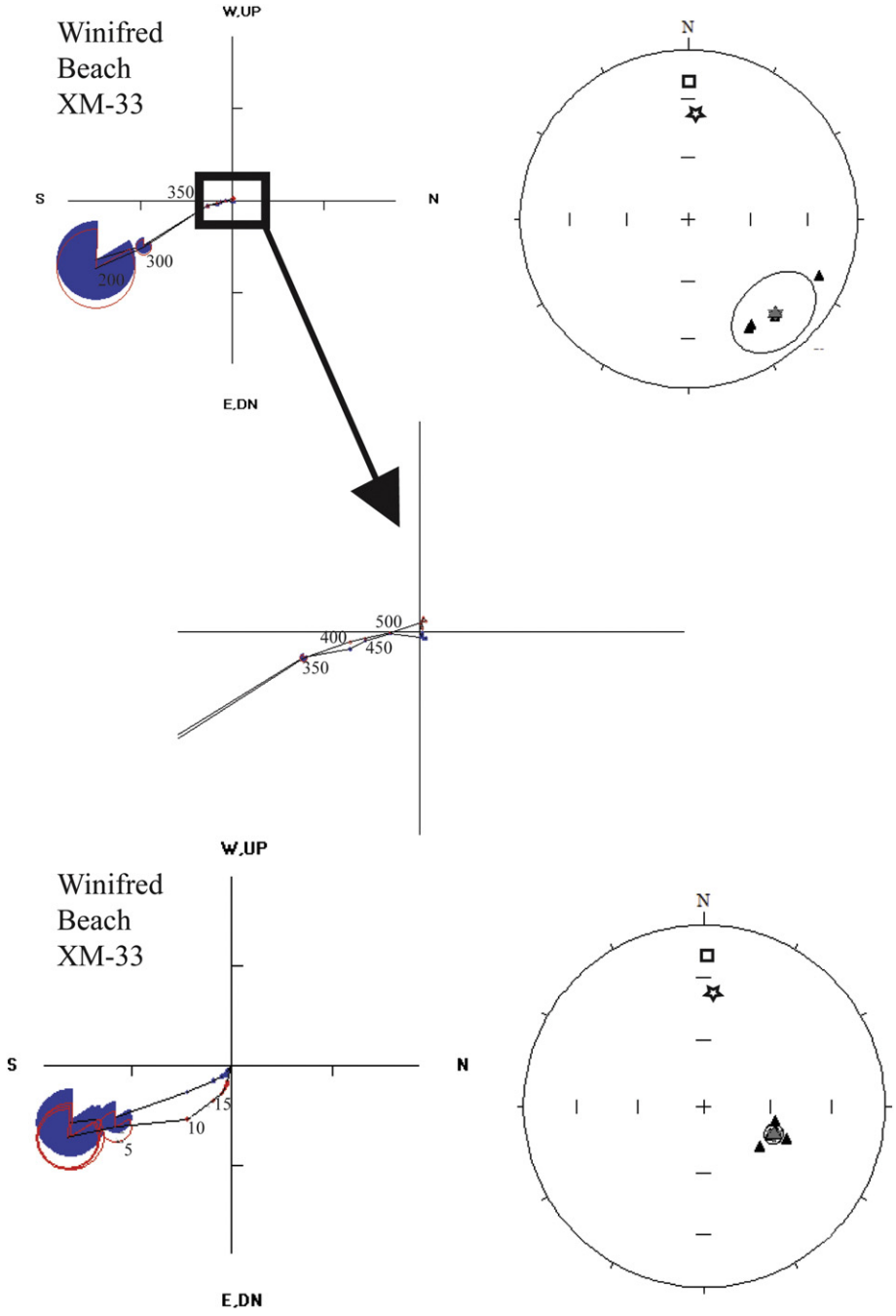


Fig. 7. Representative orthogonal vector plot (top left) and site stereonet plot (top right) and a close up of orthogonal vector diagram (middle) illustrating thermal demagnetisation level up to 570 °C for Winifred Beach. Representative orthogonal vector plot (bottom left) and site stereonet plot (bottom right) for Winifred Beach obtained using AF-demagnetisation. First tick on orthogonal vector plot represents 1000 mA/m. Blue (solid) data points are projected onto the horizontal plane, red (open) data points are projected onto vertical plane. Black star (within stereonet) is the present magnetic field, D = 1.5°, I = −39° and the Black square is the dipole field at Christmas Island.

Table 4
Summary of paleomagnetic results of four sites from Christmas Island. D_m° and I_m° represent mean declination and mean inclination respectively; α_{95}°, semi-angle of cone of 95% confidence about mean direction; k, Fishers precision parameter, n, number of specimen in a site; λ° represents the paleo-latitude; dp and dm, are the errors associated with the pole.

Site	Demag	Age (Ma)	D_m°	I_m°	α_{95}°	k	n	Paleolatitude (λ° S)	Paleolat. error		VGP Pole		Error	
									+	−	Lat°	Long°	dp	dm
Waterfall	AF	37.75 ± 0.77	8.8	−65.5	5.8	255.7	4	47.6	7.1	−8.3	52.1	275.8	7.6	9.4
Ethel	AF	37.10 ± 0.66	354.7	−57.2	23.5	28.5	3	37.8	19.4	−34.1	62.2	294.5	25.0	34.3
Dolly	AF	43.55 ± 0.45	162.2	77.6	13.1	35.0	5	66.2	22.3	−19.9	−32.9	113.9	23.0	24.5
Winifred	TH	4.32 ± 0.17	137.3	24.7	19.2	23.6	4	12.9	12.8	−10.2	−48.1	187.7	11.0	20.6

Table 5
Summary of paleomagnetic results by combining the Eocene sites from Christmas Island to determine a single pole position. Abbreviations are same as Table 4.

Site	D_m°	I_m°	α_{95}°	k	n	Paleolatitude (λ°)	Paleolat. error		Pole Lat°	Pole Long°	dp	dm
							+	−				
Waterfall + Ethel Beach	1.7	−62.2	8.3	54.5	7	−43.5	9.0	−11.2	56.9	283.2	10.00	12.9
Eocene phase	11.4	−78.2	12.9	12.2	12	−67.3	20.0	−20.3	−32.6	280.3	22.9	24.4
Mean of 3 VGP's (W + E + D)	357.4	−67.0	17.1	52.9	3	−49.7	19.0	−28.7	−50.7	288.1	23.4	28.3

the observed tilt, paleolatitude of the Pliocene phase of volcanism is between 12.03°and 12.88° S, and Waterfall and Ethel Beach combined is in the range of 41°–43° S which is still significantly larger than existing reconstruction models.

It is possible that the volcanic sites sampled were influenced by structural and tectonic forces that were not possible to determine in the field and might be responsible for such high inclination. Barrie (1967) and Polak (1976) have documented that the island has been affected by a network of faults. Trueman (1965) mentions faulting has played a major role in shaping the coastline of the island with normal faults of high angle and tens of metres of displacement being observed at many locations. Faults around the Murray Hill region have already been mentioned earlier (Polak, 1976; Baxter and Weeks, 1984). Rivereau (1965) mentions that the geomorphology of the island is influenced by terraces, faults and scarps. It is further discussed that the coastlines are affected by faulting, especially, the south-west and east-south-east, whereas the north coast is little affected by it. Two major networks of fault-lines have been reported east–west and north-north-west (Rivereau, 1965). It is conceivable that these volcanic sampling sites were influenced by faulting and associated block movements (Fig. 2), but our field observations at our sample sites suggest minimal interference.

The Indo-Australian plate has undergone major transformation in terms of plate motion since its breakup from Gondwana and its subsequent northward journey. Following the cessation of spreading in the Wharton Basin around ~43 Ma, and the initiation of spreading along the South East Indian Ridge (Liu et al., 1983 and Müller et al., 2000), the once separate Indian and Australian plates became one rigid plate. The northern boundary of the plate changes from a continental collision in the west, to a subduction zone in the east. This change in tectonic setting along the northern margin, and the relative motion of India and Australia, is causing torsional stresses within the centre of the plate (Cloetingh and Wortel, 1986). Deplus et al. (1998) discussed the distinct deformation on either side of the Ninetyeast Ridge. Towards the west, folds and east–west oriented reverse faults are common, whereas, on the east of the ridge, left lateral strike slip faulting is prevalent along a north–south trending fracture zone (Deplus et al., 1998). The deformation zone has been called as the Capricorn Plate (Royer and Gordon, 1997). Such a distinct deformation pattern within the diffuse plate boundary could have altered the tectonic fabric of the seamount province, which might influence the reconstructed paleoposition of Christmas Island. The discrepancies in the paleolatitude of the Eocene phase of Christmas Island with those from global reconstructions models (O'Neill et al., 2005; Seton et al., 2012) could possibly be related to

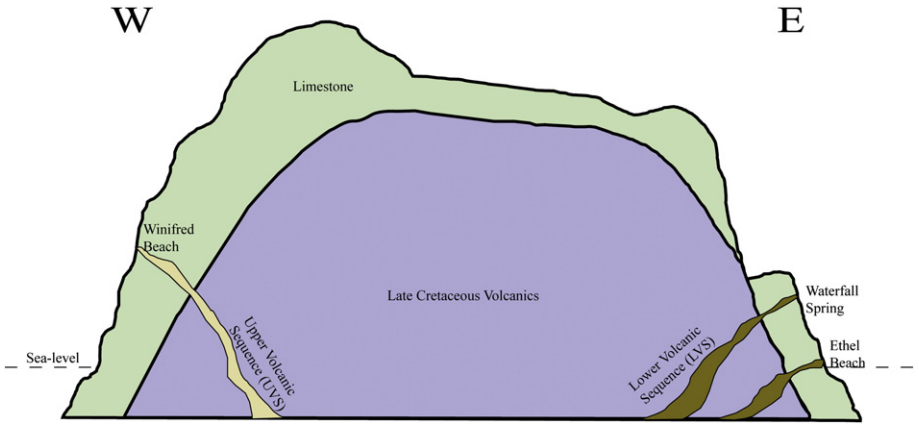


Fig. 8. A proposed diagrammatic vertically exaggerated west to east cross section of Christmas Island showing the main submerged Late Cretaceous volcanic and LVS and UVS. There is a lower volcanic sequence outcropping on the west coast at Sydney's Dale but has not been shown in this section.
Modified from Grimes (2001)

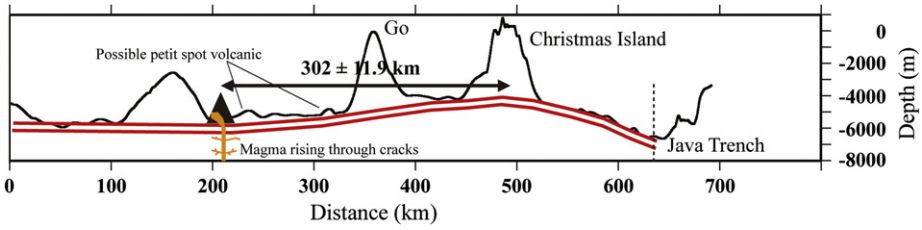


Fig. 9. Back tracking the position of Christmas Island and associated Pliocene volcanism at the seaward side of the outer rise of the subducting trench. Black line represents the profile of the oceanic lithosphere in a NE-SW section perpendicular to the trench, red line showing the bending and flexing of the lithosphere. Golden Bo'Sunbird and two small seamounts (SW of Christmas Island) represent petit spot features produced due to flexure.

Table 6
Magnetic polarity of the four sampling sites from Christmas Island and their comparison with the Geomagnetic Polarity Time Scale (Cande and Kent, 1995).

Location	Sample no.	Age (Ma)	Polarity	Magnetic chron (Cande and Kent, 1995)	Magnetic chron (Wei, 1995)
Waterfall	XM-20	37.8 ± 0.8	Normal	C17n.1n–C17n.2n–C18n.1n	C17n.1n–C17n.2n–C18n.1n
Ethel	XM-35	37.1 ± 0.7 (min age)	Normal	C16n.2r–C17n.1n–C17n.2n	C16n.2n–C17n.1n–C17n.2n
Dolly	XM-29	43.4 ± 0.5	Reverse	C20n–C20r	C20n–C20r
Winifred	XM-33	4.3 ± 0.2	Reverse	C2An.3r–C3n.2n (Gilbert)	C2An.3r–C3n.2n (Gilbert)

this deformation, and provide a further scope to constrain the paleomagnetic history of this intraplate region.

Additionally, dynamic tomography produced due to mantle flow variations and density changes can induce elevation or subsidence in continental and oceanic areas (Nyblade and Robinson, 1994; Lithgow-Bertelloni and Silver, 1998; Moucha et al., 2008). Dynamic subsidence of the order of 300 m has been reported in the Southeast Asia by Wheeler and White (2002). Variations of these natures have been observed in Southern and Eastern Africa, producing regions of elevation of the order of 1 km, that are underlain by large scale low velocity zones (Lithgow-Bertelloni and Silver, 1998). The East Indian Ocean contains three significant low velocity zones, described by Montelli et al. (2004), and suggested to be deep mantle plumes, which can cause local uplift (Hager et al., 1985; Braun, 2010). Global dynamic tomography models have been presented by Heine et al. (2008) and Steinberger et al. (2001). Heine et al. (2008) constrained present day dynamic topographic in the CHRISP between −500 to 500 m, and −100 to 500 m around the reconstructed position of Christmas Island at 70 Ma. Moucha et al. (2008) prepared global ocean dynamic topography, and constrained it within −500 to 0 m. Thus, the Northeast Indian Ocean and CHRISP have been affected by the variation in seafloor depths in response to dynamic tomography, and this could disturb the preserved tilts within the sampled units, during the progressive cooling of magmatic material, though the magnitude of this effect is likely to be similar, or less, than that ascribed to tilting due to subduction flexure.

The re-emergence of volcanism at Christmas Island in the Late Eocene is synchronous with tectonic reorganisation in the East Indian Ocean around 43 Ma (Liu et al., 1983). Plate reorganisation within the Pacific Plate is exemplified by the change in the Hawaii Emperor Bend (HEB). Dalrymple and Clague (1976) first suggested an age of 43 Ma, but Sharp and Clague (2006), using new radioisotopic $^{40}Ar/^{39}Ar$ data from additional islands revised it to 50 Ma. Tarduno (2007), however, is critical of such a value, and rather argues for an age of 47 Ma. Whittaker et al. (2007) suggested major reorganisation events correlated with HEB, such as subduction of Pacific-Izanagi spreading ridge, and initiation of Marianas and Tonga–Kermadec subduction. Such events also caused plate motion changes in the Australian plate. Slab-pull north of Australia, due to westerly subducting Wharton Ridge, led to a change in Australian plate motion, from northwest, to north, around 50 Ma (Whittaker et al., 2007). Shortly after, spreading in the Wharton Ridge ceased at 43 Ma (Liu et al., 1983, Müller et al.,

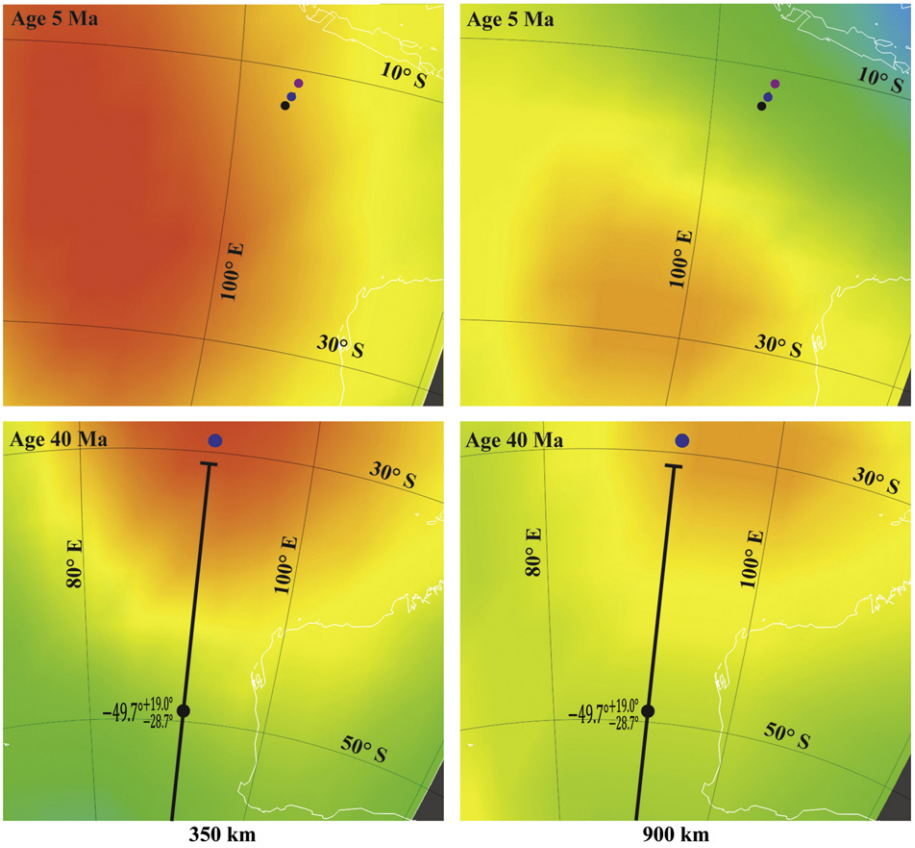


Fig. 10. Paleo-position of Christmas Island overlayed on Grand (2002) seismic tomography model at 350 and 900 km depth slices. Blue circle represents an approximate position of Christmas Island as per reconstruction model, black represents reconstructed paleo-position according to this study, and purple circle represents present position of the island. The paleo-position of the Eocene age is shown by vertical bar at 40 Ma where the circle represents the paleolatitude. This position is further south of that suggested by moving hotspot model of O'Neill et al. (2005).

2000; Singh et al., 2011). Spreading rates in the SEIR changed from 140 mm/yr to 60 mm/yr at 51.9 Ma to 45.3 Ma, respectively, when plate reorganisation took place in the Australian plate (Cande et al., 2010). Deplus et al. (1998) suggested that there are differential stress patterns on either side of the Ninetyeast Ridge, with reverse faults in the west, and left-lateral north–south strike slip faults in the east. Thus, the combined change in plate reorganisation at 50 Ma, slowing down of SEIR between 51 and 45 Ma, and closure of the Wharton Ridge, resulted in changed stress regime of now intact Indo-Australian Plate. This could have led to re-emergence of volcanism in the seamount province that has yet only been documented only at Christmas Island.

To summarise, our study shows that the position of Christmas Island can be calculated using paleomagnetic data. Our study has attempted to constrain such a position of the island with reasonable errors and found it to be further south than that predicted by the existing reconstructions model of O'Neill et al. (2005) and Seton et al. (2012). The paleoposition for the Pliocene phase of the island appears to constrain the island effectively, and is consistent with global reconstructions (Seton et al., 2012). Paleomagnetic analysis and plate reconstructions all position the island around 12–13° S at 4.3 Ma.

5. Conclusion

Volcanism in the Northeast Indian Ocean is represented in the form of submerged flat topped and rounded peak seamounts and two subaerially exposed islands, Christmas Island and Cocos (Keeling) Archipelago. Of these, Christmas Island is the only island within the region to have recorded this intraplate volcanism in the form of basaltic rocks that are presently exposed above sea-level. $^{40}Ar/^{39}Ar$ dating of the island has revealed ongoing Cenozoic volcanism, from 43 Ma to 37 Ma. This was followed by a more recent Pliocene volcanic phase at 4.32 ± 0.17 Ma. The younger volcanic event is probably related to the onset of lithospheric plate flexure, and associated asthenospheric decompression, on the seaward side of the outer rise (Fig. 9). An older event, in the Late Cretaceous, probably formed the core of island and is presently not exposed on the surface. We carried out a paleomagnetic investigation by drilling basalts at Christmas Island and have attempted to constrain island's position. The high paleolatitude of the island in the Eocene is suggestive of a more southerly position than that predicted by plate reconstruction models. This could either be due to inadequately averaged secular variation, due limited outcrop, or because of intraplate deformation within the Capricorn Plate, altering the tectonic fabric. The youngest phase, at 4.3 Ma, resolved through thermal demagnetisation, yielded a paleoposition of ~13° S which agrees well with the edge of the lithospheric bulge at that time, and with existing plate reconstruction models. It was argued by Hirano et al. (2008) that low volume intraplate volcanism could be observed at similar tectonic settings to the Japan and Tonga trench, and we put forward the Indo-Australian plate subduction setting as a likely candidate for this phase of intraoceanic volcanism. The Late Eocene event, however, is coincident with the cessation of spreading at Wharton Ridge at 43 Ma and the subsequent opening of the South East Indian Ridge. Tectonic reorganisation changed the stress regime of the once separate Indian and Australian plates, allowing deeper fertile mantle, imaged in seismic tomography, to melt, and ascend through the plate, which may have resulted in the re-activation of volcanic activity at Christmas Island between 43 and 37 Ma.

Acknowledgement

We are grateful to Trond Torsvik and an anonymous reviewer for their constructive comments that greatly improved this manuscript. The authors are thankful to the staff at Western Australian Argon Isotope Facility at Curtin University and at CSIRO, North Ryde, for helping with the analytical techniques. We also thank Christmas Island National Park staff for their time and discussions on sharing the locations of volcanic exposures at Christmas Island. This work was in part supported by Australian Research Council (ARC) funding DP110104145, DP0880801, and FT100100717. This is contribution 444 from the ARC Centre of Excellence for Core to Crust Fluid Systems (http://www.ccfs.mq.edu.au) and 936 in the GEMOC Key Centre (http://www.gemoc.mq.edu.au).

Appendix A. Supplementary data

Supplementary data to this article can be found online at http://dx.doi.org/10.1016/j.gr.2014.04.004.

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